LIMKOKWING UNIVERSITY OF CREATIVE TECHNOLOGY
SIERRA LEONE

FALCULTY OF INFORMATION AND COMMUNICATION TECNOLOGY

PROG101: PRINCIPLES OF PROGRAMMING LOGIC & DESIGN

Title : Assignment 1

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Class : BBIT(1101)

Semester/Year : 1 of 1

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1. Rationale

1.1 Problem Description

Community Health Centres (CHCs) and government hospitals in Sierra Leone commonly receive more patients than the available nurses can see at once. Patients are often attended to in order of arrival, or by the informal judgement of a busy nurse. There is no consistent, visible rule for deciding who must be seen first. As a result, a woman with a pregnancy danger sign, a child under five with a very high fever, or a patient showing signs of cholera, Lassa fever or Ebola may wait behind people with mild complaints, and infectious patients may sit among other waiting patients.

Who is affected: patients (especially pregnant women, children under five and the elderly), nurses and clinic staff who must make rapid decisions under pressure, and the wider community, which is exposed when infectious cases are not identified early. **Why the problem exists:** limited staff, high patient numbers, paper registers, the absence of a standard triage tool, unreliable electricity and internet, and limited budgets for commercial software. **How the solution helps:** MediQueue SL applies fixed, transparent triage rules to every patient, places them in a colour-coded queue, issues a ticket with an estimated waiting time, alerts staff to suspected infectious cases and produces a daily summary.

1.2 Background

Triage is the established practice of sorting patients by clinical urgency rather than arrival time (Mackway-Jones, Marsden and Windle, 2014), and the World Health Organization promotes structured emergency triage for children (World Health Organization, 2016). Since the Free Healthcare Initiative of 2010, which removed user fees for pregnant and breastfeeding women and children under five, demand on public facilities has risen. The 2014 to 2016 Ebola outbreak exposed how dangerous it is when infectious patients are not recognised early, and cholera and Lassa fever remain recurring threats (World Health Organization, n.d.a; n.d.b; n.d.c). MediQueue SL adapts the common red, yellow and green triage colours to these local conditions.

1.3 Community Impact

Problem solved: unfair and unsafe waiting at health facilities. **Expected impact:** life-threatening cases (RED) are always called first; urgent cases (YELLOW) are seen before routine cases (GREEN), while a fairness limit guarantees routine patients are not left waiting forever; suspected infectious patients are flagged on registration; and each patient receives a ticket and a wait estimate, which reduces conflict and uncertainty in the waiting area. **Beneficiaries:** patients and families, nurses in charge, facility managers (through the daily summary) and district health teams. **Long-term sustainability:** the system needs no internet and no special hardware, its thresholds (queue size, fairness limit, average consultation time) are single values that staff can adjust, and its open design allows local developers and health workers to maintain and extend it.

1.4 Proposed Solution

MediQueue SL is a menu-driven system designed in Flowgorithm. The main menu offers five options: (1) register a patient, (2) call the next patient, (3) show queue status, (4) show the daily summary and (5) exit. Registration collects the patient name, sex, age, temperature, days ill and yes/no danger-sign questions, validates every entry, assigns a

priority and issues a ticket. The design keeps all data in memory, so it does not store records after the program closes. This protects privacy in the prototype, but a future version would need secure local storage.

1.5 Computational Thinking Process

Following Wing (2006), the problem was handled through four thinking skills. **Decomposition:** the large problem was broken into 13 small modules (menu, validation, priority, queueing, calling, wait estimate, status and summary). **Pattern recognition:** the three queues (RED, YELLOW, GREEN) follow the same add, call and count pattern, and all input checks follow the same ask, test and repeat pattern. **Abstraction:** the real clinical picture was reduced to a few measurable facts (age, temperature, days ill and yes/no signs) and ten counters held in one array. **Algorithm design:** the facts are turned into precise, ordered steps using decisions and loops.

1.6 Algorithm Design

Each queue is a pair of parallel arrays (name and ticket number) with a capacity of 200, and a single array called *stats* holds all counters (for example, stats[0] is the RED count and stats[5] is the consecutive urgent calls). New patients join the back of their queue and the next patient is taken from the front, so patients in the same colour are seen first-come first-served. The priority rule uses Boolean logic: a patient is Emergency if unconscious, bleeding severely, having difficulty breathing, pregnant with a danger sign, or under 5 years with a temperature of at least 39.5; otherwise Urgent if the temperature is at least 38.0, or pregnant, or under 5, or 65 or over, or ill for 3 or more days; otherwise Routine. The fairness rule calls one GREEN patient after 4 consecutive YELLOW calls when a GREEN patient is waiting.

1.7 SDG Alignment

- **SDG 3 (Good Health and Well-being), primary:** supports faster care for emergencies and mothers and children, and early identification of infectious disease risks, in line with the Goal 3 targets on universal health coverage and on early warning and management of health risks (United Nations, 2015).
- **SDG 10 (Reduced Inequalities):** a transparent, rule-based queue treats every patient by need, not by status or connections.
- **SDG 9 (Industry, Innovation and Infrastructure):** a low-cost, offline-capable digital tool for public services.

1.8 Open Source Justification

MediQueue SL should be open source so that its triage rules can be inspected and approved by clinicians, which builds trust. Open source licences (Open Source Initiative, n.d.) remove licence fees, which matters for under-funded facilities, and let local developers adapt the system to different languages, thresholds and facility sizes. **Potential contributions:** nurses and doctors can review and refine the triage thresholds; students can translate the prompts into Krio, Mende and Temne; developers can add file or database storage, a touch-screen interface or SMS tickets; and testers can report faults through GitHub issues.

1.9 Digital Public Goods Justification

The Digital Public Goods Standard asks for relevance to the SDGs, an open licence, clear ownership, platform independence, documentation, data extraction, privacy and legal compliance, adherence to best practice and do-no-harm by design (Digital Public Goods

Alliance, n.d.). MediQueue SL can meet these by publishing its flowchart, pseudocode and README under an open licence, supporting SDG 3, storing no personal data permanently and documenting every module. It promotes **accessibility** (free, simple and usable on basic computers), **openness** (visible rules that anyone can reuse) and **public benefit** (any facility in Sierra Leone or elsewhere can adopt it without payment).

2. Algorithm (Step-by-Step)

1. **START.** Set MaxQueueSize = 200, FairnessLimit = 4, AvgConsultMinutes = 10. Create the stats array (10 counters, all set to 0) and the six queue arrays. Set running = true.
2. **WHILE** running is true: display the menu, then read a choice from 1 to 5 (repeat until valid).
3. **IF choice = 1 (Register):** read name and sex; read age (0 to 120), temperature (30.0 to 45.0) and days ill (0 to 365), each validated; ask the four yes/no danger questions; if the patient is female aged 12 to 50 ask about pregnancy and, if pregnant, about danger signs.
4. Compute the priority: 1 = Emergency (RED), 2 = Urgent (YELLOW), 3 = Routine (GREEN).
5. If that queue already holds 200 patients, output "Queue full. Refer patient to the nearest CHC." Otherwise add the patient to the end of the queue, raise the totals, raise the infectious count and output an isolation alert if signs were reported, estimate the wait (patients ahead x 10 minutes) and print the ticket.
6. **IF choice = 2 (Call next):** if all queues are empty, output "No patients are waiting." Otherwise call the first RED patient if any. If none, call the first YELLOW patient unless 4 YELLOW patients were called in a row and a GREEN patient is waiting; otherwise call the first GREEN patient and reset the consecutive count. Shift the queue forward, add 1 to the served count and output the call message.
7. **IF choice = 3 (Status):** output the count and the numbered names in the RED, YELLOW and GREEN queues.
8. **IF choice = 4 (Summary):** if no patients were registered output a message; otherwise output registered and served totals, the number and percentage (to 1 decimal place) in each category, and the suspected infectious cases.
9. **IF choice = 5 (Exit):** set running = false. When the loop ends, output "System closed. Stay safe." **END.**

3. Pseudocode

The pseudocode below is reproduced exactly from the Flowgorithm program (MediQueue_SL_Pseudocode.txt) and is shown in a smaller monospaced font so that every line of the complete system fits in the page limit.

```
Function Main
    ... MediQueue SL - triage and queue management Mohamed Turay - 905006574 (PROG101 Assignment
1)
    ... Constants (Flowgorithm has no constant statement, so they are set once here)
    Declare Integer MaxQueueSize
    Declare Integer FairnessLimit
    Declare Integer AvgConsultMinutes
    Declare Integer choice
    Declare Integer i
```

```
    Declare Boolean running

    ... stats: 0 eCount, 1 uCount, 2 rCount, 3 totalRegistered, 4 servedCount, 5 consecutiveUrgent, 6 eTotal,
7 uTotal, 8 rTotal, 9 infectiousCount
    Declare Integer Array stats[10]
    Declare String Array eName[200]
    Declare String Array uName[200]
    Declare String Array rName[200]
    Declare Integer Array eTicket[200]
    Declare Integer Array uTicket[200]
    Declare Integer Array rTicket[200]

    Assign MaxQueueSize = 200
    Assign FairnessLimit = 4
    Assign AvgConsultMinutes = 10
    For i = 0 to 9
        Assign stats[i] = 0
    End
    Assign running = true
    While running == true
        Call DisplayMenu()
        Assign choice = GetValidatedInteger("Enter choice (1-5)", 1, 5)
        If choice == 1
            Call RegisterPatient(eName, eTicket, uName, uTicket, rName, rTicket, stats, MaxQueueSize,
AvgConsultMinutes)
        Else
            If choice == 2
                Call CallNextPatient(eName, eTicket, uName, uTicket, rName, rTicket, stats, FairnessLimit)
            Else
                If choice == 3
                    Call ShowQueueStatus(eName, uName, rName, stats)
                Else
                    If choice == 4
                        Call GenerateDailySummary(stats)
                    Else
                        Assign running = false
                    End
                End
            End
        End
    End
    Output "System closed. Stay safe."
End

Function AddToQueue (Integer priority, String patientName, Integer ticketNo, String Array eName, Integer
Array eTicket, String Array uName, Integer Array uTicket, String Array rName, Integer Array rTicket, Integer
Array stats)
    If priority == 1
        Assign eName[stats[0]] = patientName
        Assign eTicket[stats[0]] = ticketNo
        Assign stats[0] = stats[0] + 1
        Assign stats[6] = stats[6] + 1
    Else
        If priority == 2
            Assign uName[stats[1]] = patientName
            Assign uTicket[stats[1]] = ticketNo
            Assign stats[1] = stats[1] + 1
```

```
            Assign stats[7] = stats[7] + 1
        Else
            Assign rName[stats[2]] = patientName
            Assign rTicket[stats[2]] = ticketNo
            Assign stats[2] = stats[2] + 1
            Assign stats[8] = stats[8] + 1
        End
    End
    Assign stats[3] = stats[3] + 1
End

Function AssignPriority (Integer age, Real temperature, Integer days, Boolean unconscious, Boolean severeBleeding, Boolean breathingDifficulty, Boolean pregnant, Boolean dangerSign)
    ... Constants: CHILD_AGE_LIMIT 5, ELDERLY_AGE_LIMIT 65, FEVER_LIMIT 38.0, HIGH_FEVER_LIMIT 39.5, LONG_ILLNESS_DAYS 3
    Declare Boolean isEmergency
    Declare Boolean isUrgent
    Declare Integer result

    Assign isEmergency = unconscious OR severeBleeding OR breathingDifficulty OR (pregnant AND dangerSign) OR (age < 5 AND temperature >= 39.5)
    Assign isUrgent = temperature >= 38.0 OR pregnant OR age < 5 OR age >= 65 OR days >= 3
    If isEmergency
        Assign result = 1
    Else
        If isUrgent
            Assign result = 2
        Else
            Assign result = 3
        End
    End
    Return Integer result
End

Function CallNextPatient (String Array eName, Integer Array eTicket, String Array uName, Integer Array uTicket, String Array rName, Integer Array rTicket, Integer Array stats, Integer fairnessLimit)
    Declare Integer i
    Declare Integer calledTicket
    Declare Integer calledPriority
    Declare String calledName

    If stats[0] + stats[1] + stats[2] == 0
        Output "No patients are waiting."
    Else
        If stats[0] > 0
            Assign calledName = eName[0]
            Assign calledTicket = eTicket[0]
            For i = 0 to stats[0] - 2
                Assign eName[i] = eName[i + 1]
                Assign eTicket[i] = eTicket[i + 1]
            End
            Assign stats[0] = stats[0] - 1
            Assign calledPriority = 1
        Else
            If stats[1] > 0 AND NOT (stats[5] >= fairnessLimit AND stats[2] > 0)
                Assign calledName = uName[0]
                Assign calledTicket = uTicket[0]
```

```
            For i = 0 to stats[1] - 2
                Assign uName[i] = uName[i + 1]
                Assign uTicket[i] = uTicket[i + 1]
            End
            Assign stats[1] = stats[1] - 1
            Assign calledPriority = 2
            Assign stats[5] = stats[5] + 1
        Else
            Assign calledName = rName[0]
            Assign calledTicket = rTicket[0]
            For i = 0 to stats[2] - 2
                Assign rName[i] = rName[i + 1]
                Assign rTicket[i] = rTicket[i + 1]
            End
            Assign stats[2] = stats[2] - 1
            Assign calledPriority = 3
            Assign stats[5] = 0
        End
    End
    Assign stats[4] = stats[4] + 1
    Output "Now calling: " & calledName & " | " & ColourName(calledPriority) & " | Ticket " & calledTicket
  End
End

Function ColourName (Integer priority)
  Declare String colour

  If priority == 1
    Assign colour = "RED"
  Else
    If priority == 2
      Assign colour = "YELLOW"
    Else
      Assign colour = "GREEN"
    End
  End
  Return String colour
End

Function DisplayMenu
  Output "========== MEDIQUEUE SL =========="
  Output "1 Register patient"
  Output "2 Call next patient"
  Output "3 Show queue status"
  Output "4 Daily summary"
  Output "5 Exit"
End

Function EstimateWaitTime (Integer priority, Integer Array stats, Integer avgMinutes)
  Declare Integer ahead
  Declare Integer result

  If priority == 1
    Assign ahead = stats[0] - 1
  Else
    If priority == 2
```

```
            Assign ahead = stats[0] + stats[1] - 1
        Else
            Assign ahead = stats[0] + stats[1] + stats[2] - 1
        End
    End
    Assign result = ahead * avgMinutes
    Return Integer result
End

Function GenerateDailySummary (Integer Array stats)
    Declare Real pct

    If stats[3] == 0
        Output "No patients registered today."
    Else
        Output "Registered: " & stats[3] & "   Served: " & stats[4]
        Assign pct = stats[6] * 100.0 / stats[3]
        Output "Emergency: " & stats[6] & " (" & Int(pct * 10 + 0.5) / 10 & "%)"
        Assign pct = stats[7] * 100.0 / stats[3]
        Output "Urgent: " & stats[7] & " (" & Int(pct * 10 + 0.5) / 10 & "%)"
        Assign pct = stats[8] * 100.0 / stats[3]
        Output "Routine: " & stats[8] & " (" & Int(pct * 10 + 0.5) / 10 & "%)"
        Output "Suspected infectious cases: " & stats[9]
    End
End

Function GetValidatedInteger (String prompt, Integer low, Integer high)
    Declare Integer value

    Do
        Output prompt
        Input value
        If value < low OR value > high
            Output "Invalid. Enter " & low & " to " & high
        End
    Loop value < low OR value > high
    Return Integer value
End

Function GetValidatedReal (String prompt, Real low, Real high)
    Declare Real value

    Do
        Output prompt
        Input value
        If value < low OR value > high
            Output "Invalid. Enter " & low & " to " & high
        End
    Loop value < low OR value > high
    Return Real value
End

Function GetYesNo (String prompt)
    Declare String answer
    Declare Boolean valid
    Declare Boolean result
```

```
    Do
        Output prompt & " (Y/N)"
        Input answer
        Assign valid = answer == "Y" OR answer == "y" OR answer == "N" OR answer == "n"
        If valid == false
            Output "Please enter Y or N"
        End
    Loop valid == false
    Assign result = answer == "Y" OR answer == "y"
    Return Boolean result
End

Function RegisterPatient (String Array eName, Integer Array eTicket, String Array uName, Integer Array uTicket, String Array rName, Integer Array rTicket, Integer Array stats, Integer maxQueue, Integer avgMinutes)
    Declare String patientName
    Declare String sex
    Declare Integer age
    Declare Integer days
    Declare Integer priority
    Declare Integer ticketNo
    Declare Integer waitMinutes
    Declare Real temperature
    Declare Boolean unconscious
    Declare Boolean severeBleeding
    Declare Boolean breathingDifficulty
    Declare Boolean suspectedInfectious
    Declare Boolean pregnant
    Declare Boolean dangerSign
    Declare Boolean queueFull

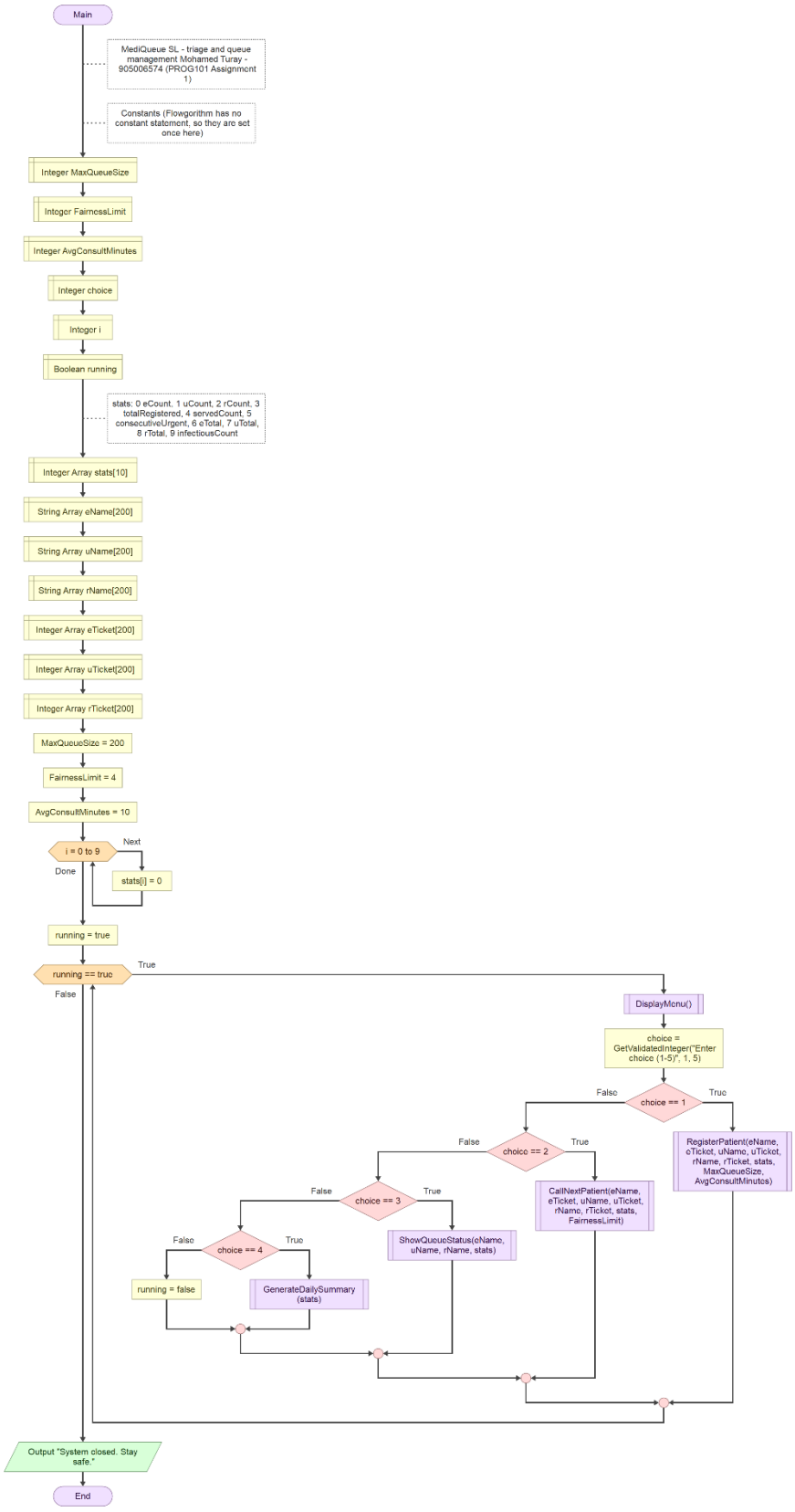
    Assign ticketNo = stats[3] + 1
    Output "Patient name:"
    Input patientName
    Output "Sex (F or M):"
    Input sex
    Assign age = GetValidatedInteger("Age in years (0-120)", 0, 120)
    Assign temperature = GetValidatedReal("Temperature in C (30.0-45.0)", 30.0, 45.0)
    Assign days = GetValidatedInteger("Days ill (0-365)", 0, 365)
    Assign unconscious = GetYesNo("Unconscious or convulsing?")
    Assign severeBleeding = GetYesNo("Severe bleeding?")
    Assign breathingDifficulty = GetYesNo("Difficulty breathing?")
    Assign suspectedInfectious = GetYesNo("Signs of cholera, Lassa fever or Ebola?")
    If (sex == "F" OR sex == "f") AND age >= 12 AND age <= 50
        Assign pregnant = GetYesNo("Pregnant?")
        If pregnant == true
            Assign dangerSign = GetYesNo("Pregnancy danger sign?")
        Else
            Assign dangerSign = false
        End
    Else
        Assign pregnant = false
        Assign dangerSign = false
    End
```

```
   Assign priority = AssignPriority(age, temperature, days, unconscious, severeBleeding, breathingDifficulty, pregnant, dangerSign)
   Assign queueFull = (priority == 1 AND stats[0] >= maxQueue) OR (priority == 2 AND stats[1] >= maxQueue) OR (priority == 3 AND stats[2] >= maxQueue)
   If queueFull == true
      Output "Queue full. Refer patient to the nearest CHC."
   Else
      Call AddToQueue(priority, patientName, ticketNo, eName, eTicket, uName, uTicket, rName, rTicket, stats)
      If suspectedInfectious == true
         Assign stats[9] = stats[9] + 1
         Output "ALERT: isolate patient and inform the nurse in charge."
      End
      Assign waitMinutes = EstimateWaitTime(priority, stats, avgMinutes)
      Output "============ MEDIQUEUE SL TICKET ============"
      Output "Ticket No: " & ticketNo & "   Name: " & patientName
      Output "Priority: " & ColourName(priority)
      Output "Estimated wait: " & waitMinutes & " minutes"
      Output "============================================="
   End
End

Function ShowQueueStatus (String Array eName, String Array uName, String Array rName, Integer Array stats)
   Declare Integer i

   Output "EMERGENCY (RED): " & stats[0]
   For i = 0 to stats[0] - 1
      Output (i + 1) & ". " & eName[i]
   End
   Output "URGENT (YELLOW): " & stats[1]
   For i = 0 to stats[1] - 1
      Output (i + 1) & ". " & uName[i]
   End
   Output "ROUTINE (GREEN): " & stats[2]
   For i = 0 to stats[2] - 1
      Output (i + 1) & ". " & rName[i]
   End
End
```

4. Flowchart



Main flowchart of MediQueue SL (Flowgorithm).

5. Functional Modules and Logical Design

Module	Purpose	Parameters / Returns
Main	Sets constants, arrays and counters; runs the menu loop and routes each choice.	None
DisplayMenu	Prints the five menu options.	None
GetValidatedInteger	Repeats (Do...Loop) until an integer is within the range.	prompt, low, high / Integer
GetValidatedReal	Same as above for decimal values such as temperature.	prompt, low, high / Real
GetYesNo	Accepts only Y, y, N or n; returns true for yes.	prompt / Boolean
RegisterPatient	Collects data, sets priority, queues the patient, shows alert and ticket.	queues, stats, maxQueue, avgMinutes
AssignPriority	Boolean rules that return 1, 2 or 3.	age, temperature, days, 5 Booleans / Integer
AddToQueue	Adds name and ticket to the correct queue and updates counters.	priority, name, ticket, queues, stats
CallNextPatient	Applies RED first, the fairness rule, shifts the queue, outputs the call.	queues, stats, fairnessLimit
EstimateWaitTime	Patients ahead x average minutes.	priority, stats, avgMinutes / Integer
ColourName	Converts 1, 2, 3 to RED, YELLOW, GREEN.	priority / String
ShowQueueStatus	Loops through each queue and prints names.	names, stats
GenerateDailySummary	Totals, percentages (1 decimal place) and infectious count.	stats

Variables, constants and data types.

Item	Type	Use
MaxQueueSize = 200, FairnessLimit = 4, AvgConsultMinutes = 10	Integer (constants)	Set once in Main because Flowgorithm has no constant statement
stats[10]	Integer Array	0 eCount, 1 uCount, 2 rCount, 3 totalRegistered, 4 servedCount, 5 consecutiveUrgent, 6 eTotal, 7 uTotal, 8 rTotal, 9 infectiousCount
eName, uName, rName [200]	String Array	Patient names in the RED, YELLOW and GREEN queues
eTicket, uTicket, rTicket [200]	Integer Array	Matching ticket numbers
temperature	Real	Body temperature in degrees Celsius
age, days, choice, priority, ticketNo	Integer	Patient and menu data
unconscious, severeBleeding, breathingDifficulty, suspectedInfectious, pregnant, dangerSign, running, queueFull	Boolean	Yes/no facts and flags used in decisions

Structures used. *Decisions:* nested If/Else for the menu, priority and queue choice. *Repetition:* a While loop for the menu, Do...Loop for validation and For loops for initialising stats, shifting queues and printing queues. *Boolean logic:* AND, OR and NOT in AssignPriority, GetYesNo and the fairness rule. *Input, processing, output:* prompts and validated input, triage and queue updates, tickets, call messages and reports.

6. Expected Outputs (Sample Scenarios)

Four patients register in this order, then the staff use options 3, 2 and 4. The results follow from the pseudocode.

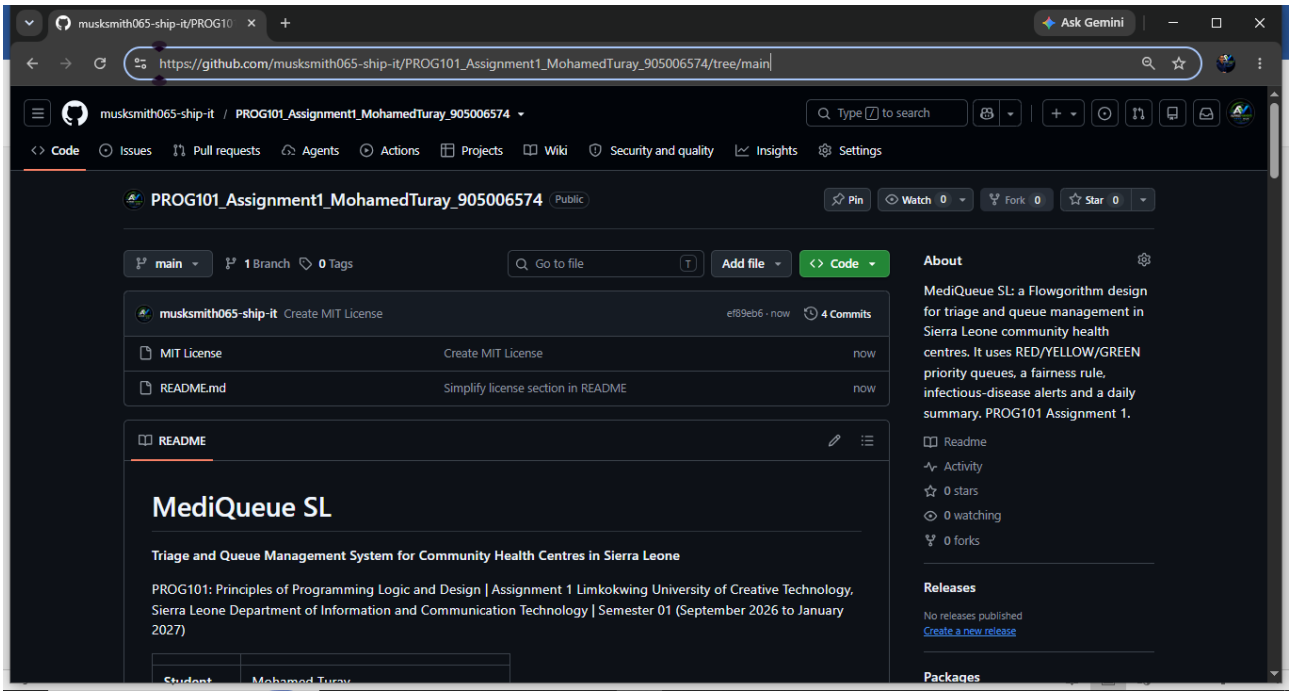
Ticket	Patient and inputs	Priority and reason	Est. wait
1	Aminata Kamara, F, 28, 37.2 C, 1 day, pregnant = Y, danger sign = Y	RED: pregnancy with danger sign	0 min
2	Ibrahim Sesay, M, 4, 38.6 C, 2 days, no signs	YELLOW: under 5 and temperature 38.0 or more (below 39.5)	10 min
3	Fatmata Bangura, F, 35, 36.8 C, 1 day, pregnant = N	GREEN: no emergency or urgent condition	20 min
4	Musa Conteh, M, 41, 39.0 C, 4 days, infectious signs = Y	YELLOW: temperature 38.0 or more; isolation alert	30 min

```
Output for ticket 4:
ALERT: isolate patient and inform the nurse in charge.
============ MEDIQUEUE SL TICKET ============
Ticket No: 4   Name: Musa Conteh
Priority: YELLOW
Estimated wait: 30 minutes
=============================================
Option 3:  EMERGENCY (RED): 1      URGENT (YELLOW): 2          ROUTINE (GREEN): 1
1.         Aminata Kamara       1. Ibrahim Sesay            1. Fatmata Bangura
2.         Musa ContehOption 2 (four times):  Now calling: Aminata Kamara | RED | Ticket 1
                   Now calling: Ibrahim Sesay | YELLOW | Ticket 2
                   Now calling: Musa Conteh | YELLOW | Ticket 4
                   Now calling: Fatmata Bangura | GREEN | Ticket 3 Option 2 (fifth time):  No patients are
waiting.
Option 4:  Registered: 4   Served: 4
         Emergency: 1 (25%)   Urgent: 2 (50%)   Routine: 1 (25%)
         Suspected infectious cases: 1
```

Validation and error cases: entering 7 at the menu or an age of 150 gives "Invalid. Enter 1 to 5" or "Invalid. Enter 0 to 120" and asks again; answering "Maybe" to a yes/no question gives "Please enter Y or N"; a full queue gives "Queue full. Refer patient to the nearest CHC."; and choosing option 4 before any registration gives "No patients registered today."

7. GitHub Evidence

Repository name: PROG101_Assignment1_MohamedTuray_905006574
Repository URL: https://github.com/musksmith065-ship-it/PROG101_Assignment1_MohamedTuray_905006574/tree/main



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